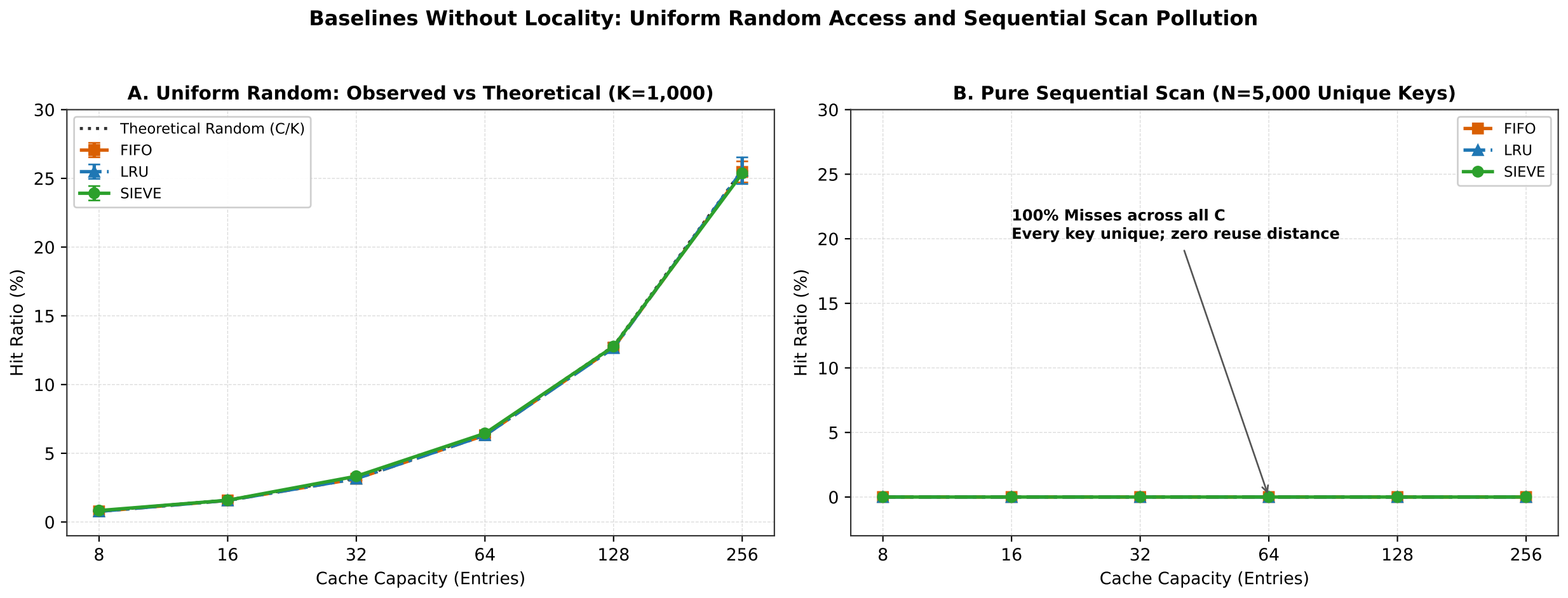
Baselines Without Locality: Uniform Random Access and Sequential Scan Pollution
A. Uniform Random: Observed vs Theoretical (K=1,000)
Theoretical Random (C/K)
FIFO
LRU
SIEVE
Hit Ratio (%)
Cache Capacity (Entries)
B. Pure Sequential Scan (N=5,000 Unique Keys)
100% Misses across all C
Every key unique; zero reuse distance
FIFO
LRU
SIEVE
Hit Ratio (%)
Cache Capacity (Entries)